

PHASE 7 OF 8

Local Deployment & Testing

How to launch LegalEase locally, the step-by-step test plan, and what the demonstration confirmed.

- Backend
- Frontend
- Test Plan
- Results

7. Local Deployment and Testing

Activity 7.1: Backend Launch

Uvicorn command (run from the project root, with the virtual environment created):

```
.\.env\Scripts\python.exe -m uvicorn backend.main:app --reload
```

Activity 7.2: Frontend Launch

Streamlit command (in a second terminal):

```
.\.env\Scripts\python.exe -m streamlit run frontend\app.py
```

Service addresses

Service	Address	Purpose
Streamlit frontend	http://localhost:8501	The user interface
FastAPI backend	http://127.0.0.1:8000	The API used by the frontend
Status check	http://127.0.0.1:8000/health	Should return {"status": "ok"}
Interactive API docs	http://127.0.0.1:8000/docs	FastAPI's built-in Swagger page

Start-up order

1. Activate the virtual environment and confirm `.env` contains your key.
2. Start the **backend** first and confirm `/health` responds.
3. Start the **frontend** in a second terminal.
4. Open `http://localhost:8501` in the browser.

Activity 7.3: Testing

The test plan follows the same path a real user takes, from opening the app to verifying the downloaded files.

#	Action	Expected result	Outcome
1	Open http://localhost:8501	LegalEase interface loads	Passed
2	Enter document details	Form accepts all six fields	Passed
3	Click Generate Document	Success message appears	Passed
4	Verify the generated preview	Structured contract shown	Passed
5	Edit the document	Editable text updates	Passed
6	Prepare TXT, DOCX and PDF	Download buttons appear	Passed
7	Verify browser downloads	All three files downloaded	Passed

The project was tested successfully on the college laboratory computer. The final demonstration confirmed generation, editing and TXT/DOCX/PDF export.

Sample request used for API testing

The same demonstration data can be sent directly to the backend (for example from the FastAPI docs page):

```
{
  "document_type": "Freelance Work Contract",
  "parties": "Arun Kumar (Freelancer), BrightTech Solutions (Client)",
  "effective_date": "2026-09-25",
  "jurisdiction": "Tamil Nadu, India",
  "terms": "Website development; total payment Rs. 50,000, 50% upfront and 50% on completion; 30-day duration; confidentiality; 15 days written notice for termination",
  "additional_instructions": ""
}
```

Recommended Automated Tests

The project includes a tests/ folder for API and exporter tests. These cases give the widest safety net for the least code:

Area	Test case	Expectation
API	GET /health	Status 200 and {"status": "ok"}
API	POST /generate with a missing required field	Status 422
API	POST /generate with the Gemini call mocked	Status 200 and a "document" key
API	Gemini call raises an exception	Status 502 with error detail
Exporters	Export a short text to DOCX and PDF	Files exist and are not empty

Example test sketch (pytest)

```
from fastapi.testclient import TestClient
from backend.main import app

client = TestClient(app)

def test_health():
    r = client.get("/health")
    assert r.status_code == 200
    assert r.json() == {"status": "ok"}

def test_generate_requires_fields():
    r = client.post("/generate", json={"document_type": "NDA"})
    assert r.status_code == 422
```

Troubleshooting

Symptom	Likely cause	What to do
Frontend cannot reach the backend	Backend not running or wrong BACKEND_URL	Start uvicorn; check the URL in .env
Browser CORS error	Frontend origin not allowed	Add the origin to the CORS list
502 error after Generate	Missing/invalid API key, network or model problem	Check GEMINI_API_KEY, GEMINI_MODEL and internet access
PDF shows a box for ₹	Built-in font lacks the glyph	Register a Unicode TTF font in ReportLab